

Haocheng He

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Professional Summary

Aviation Engineering MSc candidate at PolyU (GPA: 3.95/4.3), with an IE undergraduate and Law minor from GDUT. Experience spans airport simulation (AnyLogic, 5 routing algorithms), manufacturing optimization (line balance 58% → 70%), and Ascend 310B hardware R&D. 3 patent applications.

Education

The Hong Kong Polytechnic University Master of Science in Aviation Engineering and Operations Management GPA: 3.95 / 4.3 (Current) Thesis: Simulation Modelling and Optimization for Airport Ferry Vehicle Scheduling under Uncertainty	Hong Kong SAR, China Sep 2025 – Jan 2027
Guangdong University of Technology Bachelor of Engineering in Industrial Engineering GPA: 3.67 (Top 20%) Thesis: AnyLogic-Based Simulation Optimization of Emergency Vehicle Route Planning in Large Airports Honors: University-level Excellent Graduation Thesis, Graduation Design Innovation Award	Guangzhou, China Sep 2021 – Jun 2025
Bachelor of Laws (Minor Degree in Law) GPA: 3.61 Thesis: Coordination Between Copyright Protection and Privacy Protection in Digital Rights Management	Sep 2022 – Jun 2025

Experience

Assistant AI Engineer <i>Shenzhen Wuxian Hongyin Tech.</i> - Participated in Ascend 310B smart piano R&D across 3 development phases; tested 10+ boards (CPLD, STM32), iterated motor driver 3 times, developed firmware and authored 5 technical documents. - Participated in camera- and Ascend-based piano fingering recognition module integration (Python/C++); diagnosed cross-layer issues. - Maintained Ubuntu servers, configured Huawei network equipment, operated 3D printer, supported technical interviews.	May 2025 – Jun 2026 Shenzhen, China
Research Assistant <i>GDUT (with Guangzhou Baiyun Int'l Airport)</i> - Built an airport road-network, apron, and runway-area simulation model in AnyLogic, Java, and MySQL for aircraft landing and emergency fire-truck dispatch scenarios. - Designed single/multi-vehicle, single/multi-event, and runway-road-closure scenarios; compared D* Lite, A*, Dijkstra, BFS, and GA routing algorithms. - Completed 4 emergency-scenario comparisons across 5 routing algorithms; D* Lite achieved the highest average reliability index (1.342).	Sep 2023 – Jun 2025 Guangzhou, China
Industrial Engineering Intern <i>Midea Group — Kitchen & Water Heater Division</i> - Conducted stopwatch time studies across all workstations on a dishwasher sheet-metal line, built operator workload tables, and identified 5 over-cycle steps. - Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.	Apr 2024 – May 2024 Foshan, China
Manufacturing Operations Intern <i>Guangxi LiuGong Machinery Co., Ltd.</i> - Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis. - Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation.	Jul 2023 – Aug 2023 Liuzhou, China

Projects

Aircraft Engine Remaining Useful Life Prediction (Course Project) <i>Guangdong University of Technology</i> - Built an RUL prediction workflow on NASA C-MAPSS FD001 multivariate time-series data; cleaned operating-condition data and 21 sensor signals, constructed sequence features, and applied PCA. - Compared LSTM, BiLSTM, Transformer, and Random Forest baselines for time-series forecasting; after iterative tuning, the BiLSTM model reached about 8% final error from an initial model error of about 50%.	Apr 2024 – Jul 2024 Guangzhou
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Skills & Intellectual Property

Programming: Python, Java, SQL/MySQL, C (Embedded/STM32), JS/TS
ML & Algorithms: LSTM, BiLSTM, Transformer, Random Forest, PCA, Dijkstra, A*, BFS, GA, OR-Tools
Engineering: AnyLogic, Plant Simulation, Ubuntu/Linux, Git, EDA, Oscilloscope, Logic Analyzer
AI: LLM Workflows, Prompt Engineering
Languages: English (IELTS 6.5), Mandarin (native), Cantonese (conversational)
Patent Applications: <i>MCMC Motion Control for PI Coating Machine Controller</i> (CN122085794A, 2026); <i>Coating Control via Deep Reinforcement Learning</i> (CN121325552A, 2025); <i>Energy Regulation for Household Solar Storage</i> (CN117767437A, 2024). UM: <i>Wire Harness Production Equipment</i> (CN221352455U, 2024). SW: <i>Financial Data Analysis</i> (2024SR0146827); <i>Flood Dispatch</i> (2024SR0549247).